Finding Routing — dedup, rate-limit, and reviewer-UI dispatch

SPEC-SW-002 v0.1 DRAFT Jeremie Nunez

1 Context

The nano-sweep producer (SPEC-SW-001) can emit hundreds of draft findings per run. Raw throughput would flood the reviewer UI and generate duplicate entries across successive runs. This spec defines the routing layer that sits between the pending buffer and the reviewer: it deduplicates overlapping findings, rate-limits notifications, decides which audience (role / tier) receives which source, and wires sweep completions into the messaging channel. It does not decide the *truth* of a finding — only who sees it and when.

2 Goals

- Map each finding source (sweep, research cycle, cortex output) to a set of audience slugs.
- Deduplicate findings whose (category, target, title) triple matches an existing pending entry within a TTL window.
- Rate-limit reviewer notifications so a single sweep produces at most one digest per audience.
- Dispatch a summary message to reviewers on sweep completion, idempotently.

3 Non-Goals

Non-Goal

This spec does not generate findings (see SPEC-SW-001) nor apply resolutions (see SPEC-SW-003). It does not implement the reviewer UI itself — only the server-side dispatch logic that feeds it.

4 Invariants

Invariant

The routing layer is read-only with respect to source-of-truth tables. It only reads the pending buffer and writes to the messaging / inbox channel.

Invariant

A single sweep completion produces at most one notification per audience slug, regardless of how many findings it contains.

Invariant

Two findings with identical `(category, targetKey, normalizedTitle)` within the same TTL window MUST NOT both reach the reviewer inbox. The later one is suppressed or merged.

Invariant

Administrative reviewers always receive every source via the admin dispatch path; tier maps explicitly exclude them to avoid double delivery.

5 Interface

```
export type InboxSource =
  | "finding" | "sweep" | "research_cycle" | "consumption";

// Cortex-emitted findings: map cortex id -> audience slugs
export function getTiersForCortex(cortex: string): string[];

// Non-cortex sources: map source -> audience slugs
export function getTiersForSource(source: InboxSource): string[];

// Wire sweep producer -> reviewer inbox
export function wireSweepNotifications(
  sweepService: NanoSweepService,
  deps: {
    findAdminIds: () => Promise<string[]>;
    messagingService: Pick<MessagingService, "send">;
  },
): void;
```

6 Scenarios

Scenario: Sweep completion notifies admins

Given `wireSweepNotifications` has been called once at startup
When a sweep completes with 42 suggestions stored
Then exactly one inbox message is sent
And its recipients are the full admin id list
And its body reports group count, suggestion count, cost and duration.

Scenario: No admins configured

Given `findAdminIds()` resolves to an empty array
When a sweep completes
Then no inbox message is sent
And the sweep run still succeeds and returns its `SweepResult`.

Scenario: Deduplication within TTL

Given finding F with key $(cat, target, title)$ already sits in the pending buffer, unreviewed
When a new run emits F' with the same key within the TTL window
Then F' is suppressed (or merged into F with incremented count)
And the reviewer still sees a single row for that key.

Scenario: Rate-limit per audience

Given two sweep completions fire within the rate-limit window
When both would target the same audience

Then only the first dispatches a notification
And the second is dropped or collapsed into the first digest.

Scenario: Tier-specific cortex finding

Given a cortex whose tier map lists `["tier-b"]`
When a finding from that cortex enters the router
Then tier-b reviewers receive it
And no other non-admin tier receives it
And admins still receive it via the admin dispatch path.

7 Acceptance Criteria

Acceptance Criterion AC-1

`getTiersForSource("sweep")` returns an empty array (admin-only path). The admin path is exercised separately in the wiring.

Acceptance Criterion AC-2

`wireSweepNotifications` registers exactly one `onComplete` handler on the sweep service.

Acceptance Criterion AC-3

When a sweep completes and at least one admin id exists, `messagingService.send` is called exactly once with those recipients.

Acceptance Criterion AC-4

When a sweep completes and admin id list is empty, `messagingService.send` is not called.

Acceptance Criterion AC-5

Given two findings with identical `(category, targetKey, normalizedTitle)` submitted within the dedup TTL, the pending buffer contains exactly one reviewer-visible entry for that key.

Acceptance Criterion AC-6

Given two sweep completions within the rate-limit window, at most one notification per audience is dispatched.

Acceptance Criterion AC-7

For any cortex not present in the tier map, `getTiersForCortex` returns an empty array (fail-closed).

8 Traceability

AC	Test file	Test name
AC-1	`tests/unit/finding-routing.sources.spec.ts`	`sweep source maps to admin-only`
AC-2	`tests/unit/finding-routing.wiring.spec.ts`	`registers one completion handler`
AC-3	`tests/integration/finding-routing.notify.spec.ts`	`dispatches once to all admins`
AC-4	`tests/unit/finding-routing.wiring.spec.ts`	`skips send when no admins`
AC-5	`tests/integration/finding-routing.dedup.spec.ts`	`dedups identical finding keys`
AC-6	`tests/integration/finding-routing.ratelimit.spec.ts`	`limits digests per window`
AC-7	`tests/unit/finding-routing.cortex.spec.ts`	`unknown cortex fails closed`

9 Open Questions

- Dedup key: exact-match only, or include a lightweight similarity threshold on title?
- Rate-limit window duration — per-run, per-hour, per-day?
- Should merged duplicates carry a visible *seen N times* counter in the reviewer row?